

English Language
for *Pharmacy*
Faculty Students
First Year

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Introduction

Comprehensive Pharmacy Summary

1. History of Pharmacy

Pharmacy is one of the oldest professions in human history. People began using plants, minerals, and animal products to treat diseases thousands of years ago. Ancient civilizations like the Egyptians, Chinese, and Mesopotamians recorded medical recipes on papyrus and tablets. Greek physicians such as Hippocrates and Dioscorides organized early knowledge of herbs and medicines. Over time, pharmacy separated from medicine and became a specialized field focused on preparing, compounding, and dispensing medications.

2. Development of Pharmacy Among Arabs and Famous Arab Pharmacists

The Arabs made great contributions to pharmacy during the Islamic Golden Age (8th–14th centuries). They built the first true pharmacies (Bayt al-Saydala) and introduced standards for drug quality. They translated medical books from Greek, Persian, and Indian sources.

Famous Arab pharmacists:

Al-Razi (Rhazes): Wrote medical encyclopedias and introduced many new treatments.

Ibn Sina (Avicenna): His book The Canon of Medicine guided pharmacy and medicine in Europe for centuries.

Al-Biruni: Wrote about drugs, weights, and measures.

Ibn al-Baytar: Compiled a large book on medicinal plants.

3. Development of Pharmacy in the West

Western pharmacy grew during the Middle Ages and Renaissance. Apothecaries prepared medicines, and universities began teaching pharmacy. In the 19th century, chemistry advanced rapidly, leading to the discovery of active substances like morphine and aspirin. The 20th century introduced modern pharmaceutical companies, antibiotics, vaccines, and regulatory laws. Today, Western pharmacy includes clinical pharmacy, industrial pharmacy, and pharmaceutical research.

4. Old Methods Used to Treat Diseases

Before modern medicine, people used many traditional methods:

علاجات أعشاب
Herbal remedies

علاجات روحانية
Spiritual healing and prayers

الحجامة
Bloodletting

علاجات
Use of hot and cold therapies

حيوان
Animal products (milk, honey, fats)

كبريت
Mineral-based medicines (sulfur, salts) These methods often depended on cultural beliefs and trial-and-error. خطأ
تجريبية

5. Herbs and Pharmacy

أساسي
Herbal medicine is the foundation of pharmacy. Many drugs came from plants, such as:

سواء الصفصاف
Aspirin (from willow bark)

خشب الاصفر

Morphine (from opium poppy)

Quinine (from cinchona bark) Pharmacognosy is the science of studying medicinal plants. Today, herbs are used in complementary and alternative medicine, but modern pharmacy relies on tested, standardized drugs.

الطب البديل
تجارب
معايير

6. Pharmaceutical Education and Its Development in the Arab World

Pharmacy education in Arab countries has developed greatly. Many universities established colleges of pharmacy in the 20th century. The curriculum now includes:

منهج

Pharmacology

Pharmaceutics

Clinical pharmacy

Pharmaceutical chemistry

توسيع

آماله

Pharmacy practice Arab countries are expanding training programs, internships, and continuous professional development to match global standards.

مكتبي

كاسي

7. Drug Manufacturing •

مرادف

Drug manufacturing involves several stages:

خبرستان

Research and discovery of new chemical or biological molecules.

Pre-clinical testing in laboratories.

Clinical trials on humans.

مراقبة
Production in industrial facilities.

Quality control to check safety and purity.

تغليف وتوزيع
Packaging and distribution. Pharmaceutical industries must follow Good Manufacturing Practices (GMP).

8. Pharmacy Profession and Professional Competence

أمانة مهنية
A pharmacist must have strong knowledge and skills, including:

Understanding drugs and their effects

مراقبة دقيقة
Accurate dispensing

Communication with patients

Problem-solving abilities

أخلاق
Ethical decision-making. Continuing education ensures that pharmacists stay updated with new medications and technologies.

9. Ethics of the Pharmacy Profession

Pharmacy ethics include:

نزاهة
Honesty and integrity

احترام خصوصية
Respecting patient privacy

Providing correct information

أدوية
Avoiding harm

سلوك
Professional behavior A pharmacist must always put the patient's health and safety first.

10. The Role of the Pharmacist in the Community

Pharmacists play an important role in healthcare:

Dispensing medications

ارشاد
Counseling patients on proper use

مراقبة تفاعلات
Monitoring drug interactions

لقاح
Giving vaccinations (in many countries)

Educating the community about diseases

Supporting public health programs

11. Relationship Between the Patient and the Pharmacist

A good relationship is based on:

Trust and respect

Clear communication

الموثوقية
Confidentiality

واضحة
الالتزام بالدواء
Providing understandable instructions A strong relationship improves medication adherence and health outcomes.

12. The Future of Pharmacy

Pharmacy is changing with new technologies:

أطبب الشخصي

Personalized medicine

Pharmacogenomics

Artificial intelligence in drug discovery

Telepharmacy

Advanced clinical roles, Future pharmacists will work more closely with physicians and healthcare teams.

13. 100 Pharmacy Terms (with Arabic Translation)

Drug — دواء	Dose — جرعة
Dispensing — صرف الدواء	Prescription — وصفة طبية
OTC (Over-the-Counter) — نواء بدون وصفة	Antibiotic — مضاد حيوي
Analgesic — مسكن ألم	Antipyretic — خافض حرارة
Antiseptic — مطهر	Antiviral — مضاد فيروسات
Antifungal — مضاد فطريات	Side effect — أثر جانبي
Adverse reaction — تفاعل ضار	Contraindication — موانع استعمال
Pharmacology — علم الادوية	Pharmacokinetics — حركية الدواء
Pharmacodynamics — ديناميكية الدواء	Absorption — امتصاص

Distribution — توزيع	Metabolism — أيض / استقلاب
Excretion — إخراج	Bioavailability — التوافر الحيوي
Drug Interaction — تفاعل دوائي	Therapeutic effect — تأثير علاجي
Toxicity — سمية	Tablet — قرص

Capsule — كبسولة	Syrup — شراب
Suspension — معلق حساء شراب	Injection — حقن
Infusion — تسريب	Ointment — مرهم
Cream — كريم	Gel — جل
Inhaler — بخاخ	Nebulizer — جهاز رذاذ
Vaccine — لقاح	Immunity — مناعة
Generic drug — دواء جنيس	Brand drug — دواء تجاري
FDA — هيئة الغذاء والدواء	GMP — ممارسات التصنيع الجيد
Clinical trial — تجربة سريرية	Compounding — تركيب الأدوية
Dispensing error — خطأ صرف الدواء	Medication adherence — الالتزام الدوائي
Counseling — إرشاد دوائي	Dosage form — شكل جرعي
Controlled drug — دواء مراقب	Pharmacy practice — ممارسة الصيدلة
Pharmacist — صيدلي	Pharmacy technician — مساعد صيدلي
Prescription label — ملصق الوصفة	Storage conditions — شروط التخزين
Refill — إعادة صرف	Shelf life — مدة صلاحية
Expiry date — تاريخ انتهاء	Dose calculation — حساب الجرعة
Parenteral — خارج الجهاز الهضمي	Oral route — طريق فموي

Topical route — طريق موضعي	Subcutaneous — تحت الجلد
Intravenous — وريدي	Intramuscular — عضلي
Tolerance — تحفل	Dependence — اعتماد
Addiction — إدمان	Poisoning — تسمم
Antidote — ترياق	Herbal medicine — طب الأعشاب
Pharmacognosy — علم العقاقير	Toxicology — علم السموم
Formulation — صياغة دوائية	Stability — ثباتية
Solubility — ذوبانية	Emulsion — مستحلب

Dilution — تخفيف	Concentration — تركيز
Sterile products — مستحضرات معقمة	Aseptic technique — تقنية تعقيم
Biopharmaceutics — علم الأحياء الدوائي	Biotechnology — التكنولوجيا الحيوية
Nanotechnology — تقنية النانو	Drug delivery — إيصال الدواء
Quality control — ضبط الجودة	Regulatory affairs — الشؤون التنظيمية
Batch number — رقم التشغيل	Clinical pharmacy — صيدلة سريرية
Community pharmacy — صيدلة المجتمع	Hospital pharmacy — صيدلة المستشفيات
Industrial pharmacy — صيدلة صناعية	Evidence-based medicine — الطب المبني على الدليل
Patient profile — ملف المريض	Drug therapy plan — خطة العلاج الدوائي
Monitoring — مراقبة	Compliance — التزام
Counseling room — غرفة الإرشاد	Documentation — توثيق
Public health — الصحة العامة	Universal precautions — احتياطات عامة
Diagnosis — تشخيص	Symptom — عرض
Sign — علامة	Treatment — علاج
Therapy — علاج/معالجة	Acute — حاد

Chronic — مزمن	Infection — عدوى
Inflammation — التهاب	Fever — حمى
Pain — ألم	Fatigue — تعب
Nausea — غثيان	Vomiting — قيء
Allergy — حساسية	Asthma — ربو
Diabetes — سكري	Hypertension — ارتفاع ضغط الدم
Stroke — جلطة دماغية	Heart attack — نوبة قلبية
Cancer — سرطان	Tumor — ورم
Virus — فيروس	Bacteria — بكتيريا
Parasite — طفيل	Immune system — جهاز المناعة
Blood pressure — ضغط الدم	Pulse — نبض
Respiration — تنفس	X-ray — أشعة سينية
MRI — تصوير بالرنين المغناطيسي	CT scan — تصوير مقطعي محوسب
Ultrasound — موجات صوتية	Surgery — جراحة

Hypertension
ارتفاع ضغط
الدم

Operation — عملية	Anesthesia — تخدير
Fracture — كسر	Antibiotic resistance — مقاومة المضادات
Wound — جرح	Bleeding — نزيف
Dehydration — جفاف	Hydration — ترطيب
Obesity — سمنة	Malnutrition — سوء تغذية
Infection control — مكافحة العدوى	Hygiene — النظافة
Diet — حمية	Nutrition — تغذية
Hormone — هرمون	Thyroid — الغدة الدرقية
Kidney — كلية	Liver — كبد
Lung — رئة	Brain — دماغ
Heart — قلب	Skin — جلد

Muscle — عضلة	Bone — عظام
Joint — مفصل	Nervous system — الجهاز العصبي
Digestive system — الجهاز الهضمي	Respiratory system — الجهاز التنفسي
Cardiovascular system — الجهاز القلبي الوعائي	Endocrine system — الجهاز الغدي
Urinary system — الجهاز البولي	Reproductive system — الجهاز التناسلي
Blood test — فحص دم	Urine test — فحص بول
Biopsy — خزعة كبيحة	Swelling — تورم
Rash — طفح جلدي	Cough — سعال
Cold — نزلة برد	Flu — إنفلونزا
Headache — صداع	Migraine — شقيقة
Depression — اكتئاب	Anxiety — قلق
Insomnia — أرق	Seizure — نوبة تشنج
Trauma — صدمة	Poisoning — تسمم
Burn — حرق	Infection spread — انتشار العدوى
Quarantine — حجر صحي	Prevention — وقاية
Vaccination — تطعيم	First aid — إسعافات أولية
CPR — إنعاش قلبي رئوي	Emergency — طوارئ
Intensive care — عناية مركزة	Monitoring — مراقبة
	Vital signs — العلامات الحيوية
	Medication list — قائمة الأدوية
Medical history — التاريخ الطبي	Prognosis — توقع سير المرض
Follow-up — متابعة	Rehabilitation — تأهيل
Recovery — تعاف	
Healthy lifestyle — أسلوب حياة صحي	

Medical Terms (With Arabic Translation)

Body organs — أعضاء الجسم		
Heart — القلب	Brain — الدماغ	Lung — الرئة
Stomach — المعدة	Kidney — الكلية	Stomach — المعدة
Bladder — المثانة	Pancreas — البنكرياس	Intestine — الأمعاء
Spleen — الطحال	Esophagus — المريء	Gallbladder — المرارة
Muscle — العضلة	Skin — الجلد	Bone — العظم
Spinal cord — الحبل الشوكي	Thyroid gland — الغدة الدرقية	Adrenal gland — الغدة الكظرية
Ovary — المبيض	Liver — الكبد	Testis — الخصية
Kidney — الكلية		
Diseases — الأمراض		
Diabetes mellitus — داء السكري	Hypertension — ارتفاع ضغط الدم	Asthma — الربو
Anemia — فقر الدم	Cancer — السرطان	Tuberculosis — السل
Hepatitis — التهاب الكبد	Pneumonia — الالتهاب الرئوي	Arthritis — التهاب المفاصل
Osteoporosis — هشاشة العظام	Stroke — السكتة الدماغية	Heart failure — فشل القلب
Peptic ulcer — القرحة الهضمية	Kidney failure — الفشل الكلوي	Infection — عدوى
Migraine — الشقيقة (الصداع النصفي)	Epilepsy — الصرع	Depression — الاكتئاب
Obesity — السمنة	Allergy — الحساسية	
Medical Devices — الأجهزة الطبية		
Stethoscope — سماعة الطبيب	Thermometer — ميزان الحرارة	Blood pressure monitor — جهاز قياس ضغط الدم
Glucometer — جهاز قياس السكر	ECG machine — جهاز تخطيط القلب	X-ray machine — جهاز الأشعة السينية
MRI scanner — جهاز الرنين المغناطيسي	CT scanner — جهاز الأشعة المقطعية	Ultrasound machine — جهاز الموجات فوق الصوتية
Ventilator — جهاز التنفس الصناعي	Defibrillator — جهاز إزالة الرجفان	Infusion pump — مضخة المحاليل
Dialysis machine — جهاز غسيل الكلى	Nebulizer — جهاز الاستنشاق	Pulse oximeter — جهاز قياس الأكسجين
Diagnostic Tests — الفحوصات التشخيصية		

Blood test — تحليل الدم
Stool analysis — تحليل البراز
Lipid profile — تحليل الدهون
Biopsy — خزعة

cholera are available and are used in high-risk areas to control outbreaks. Public health education plays a major role in reducing the burden of cholera worldwide.

Asthma

Asthma is a chronic respiratory disease characterized by inflammation and narrowing of the airways. It affects people of all ages, but it often begins in childhood. Asthma causes recurring episodes of wheezing, shortness of breath, chest tightness, and coughing, especially at night or early in the morning.

The exact cause of asthma is not fully understood, but it is believed to result from a combination of genetic and environmental factors. Common triggers include allergens such as dust mites, pollen, animal dander, air pollution, smoke, respiratory infections, cold air, and physical exercise. When a person with asthma is exposed to a trigger, the airway muscles tighten, the lining becomes swollen, and excess mucus is produced, making breathing difficult.

Diagnosis of asthma is based on medical history, physical examination, and lung function tests such as spirometry. These tests measure how much air a person can breathe in and out and how fast. Sometimes, allergy tests are also performed to identify specific triggers.

Although asthma has no cure, it can be effectively controlled with proper treatment. Medications are divided into two main groups: relievers and controllers. Reliever medications, such as short-acting bronchodilators, provide quick relief during asthma attacks. Controller medications, including inhaled corticosteroids, help reduce airway inflammation and prevent symptoms in the long term. Patients are advised to use inhalers correctly and follow an asthma action plan.

Preventing asthma attacks involves avoiding known triggers, maintaining a clean environment, and taking medications as prescribed. With proper management, most people with asthma can lead normal, active lives.

Diabetes Mellitus

Diabetes mellitus is a chronic metabolic disorder characterized by high levels of glucose in the blood. It occurs either when the body does not produce enough insulin or when the body cannot effectively use the insulin it produces. Insulin is a hormone that helps glucose enter the cells to be used as energy. There are two main types of diabetes. Type 1 diabetes usually develops in childhood or adolescence and is caused by the immune system destroying insulin-producing cells in the pancreas. Patients with type 1 diabetes require lifelong insulin therapy. Type 2 diabetes is more common and is often associated with obesity, physical inactivity, and genetic factors. In this type, the body becomes resistant to insulin.

Common symptoms of diabetes include frequent urination, excessive thirst, increased hunger, unexplained weight loss, fatigue, and slow wound healing. If diabetes is not well controlled, it can lead to serious complications such as heart disease, kidney failure, nerve damage, vision problems, and infections.

Reading

اعراض Malaria: Causes, Symptoms, Treatment, and Prevention

مضغ
مضغ
مضغ
Malaria is a serious infectious disease caused by Plasmodium parasites, which are transmitted to humans through the bite of infected female Anopheles mosquitoes. It remains one of the world's major public health challenges, especially in tropical and subtropical regions such as sub-Saharan Africa, parts of Asia, and South America. Despite global efforts to reduce its spread, malaria continues to cause millions of infections and hundreds of thousands of deaths each year.

انواع
There are several species of Plasmodium that can infect humans, but the most severe form is caused by Plasmodium falciparum. Other species, such as Plasmodium vivax and Plasmodium malariae, also cause illness but are generally less dangerous. Once an infected mosquito bites a person, the parasites enter the bloodstream and travel to the liver, where they multiply. Later, they re-enter the bloodstream and infect red blood cells, leading to the symptoms associated with the disease.

قشعريرة
Malaria symptoms usually appear 10 to 15 days after infection. The most common symptoms include high fever, chills, sweating, headache, fatigue, and muscle pain. Some patients may experience nausea, vomiting, and diarrhea. In severe cases, malaria can cause anemia, jaundice, seizures, kidney failure, or even coma. Without prompt treatment, severe malaria—particularly that caused by Plasmodium falciparum—can be fatal.

وجود
تشخيص
Diagnosis is typically made through blood tests that detect the presence of the parasite. Early diagnosis is crucial because it greatly improves the chances of recovery. Treatment depends on the type of parasite and the severity of the infection. The most widely used medicines include artemisinin-based combination therapies (ACTs), which are considered the most effective treatment for P. falciparum malaria. In areas where drug resistance exists, different medications or combinations may be used.

Blood test — تحليل الدم	Complete blood count (CBC) — صورة الدم الكاملة	Urine analysis — تحليل البول
Stool analysis — تحليل البراز	Liver function test (LFT) — اختبار وظائف الكبد	Blood glucose test — تحليل سكر الدم
Lipid profile — تحليل الدهون	Thyroid function test — اختبار وظائف الغدة الدرقية	Pregnancy test — اختبار الحمل
Biopsy — خزعة	Culture and sensitivity — مزرعة وحساسية	PCR test — اختبار تفاعل البوليميراز المتسلسل
Serology test — الفحوصات المصلية	Rapid test — الفحص السريع	
Common Medical Terms — مصطلحات طبية عامة		
Diagnosis — التشخيص	Treatment — العلاج	Symptoms — الأعراض
Side effects — الآثار الجانبية	Dosage — الجرعة	Prescription — وصفة طبية
Antibiotic — مضاد حيوي	Analgesic — مسكن للألم	Anti-inflammatory — مضاد للالتهاب
Vaccine — لقاح	Infection control — مكافحة العدوى	Sterilization — التعقيم
Immunity — المناعة	Chronic disease — مرض مزمن	Acute disease — مرض حاد
Pathology — علم الأمراض	Pharmacology — علم الأدوية	Toxicology — علم السموم
Clinical trial — تجربة سريرية	Patient history — التاريخ المرضي	
Additional Terms — مصطلحات إضافية		
Vital signs — العلامات الحيوية	Blood transfusion — نقل الدم	Oxygen therapy — العلاج بالأكسجين
IV injection — حقن وريدي	Intramuscular injection — حقن عضلي	Subcutaneous injection — حقن تحت الجلد
Emergency — حالة طوارئ	Intensive Care Unit (ICU) — وحدة العناية المركزة	
Outpatient — مريض خارجي	Inpatient — مريض داخلي	

Typhoid Fever

Typhoid fever is a serious infectious disease caused by the bacterium *Salmonella typhi*. It spreads mainly through contaminated food and water, especially in areas where sanitation and clean water supplies are limited. The disease remains a major public-health concern in many developing countries, affecting millions of people each year.

After a person becomes infected, symptoms usually appear within 7 to 14 days. The most common symptoms include prolonged high fever, fatigue, weakness, headache, and abdominal pain. Many patients also experience loss of appetite, diarrhea, or constipation. In some cases, a faint rash of flat, rose-colored spots may appear on the chest and abdomen. If left untreated, typhoid fever can lead to serious complications such as intestinal bleeding, perforation, and even death.

Diagnosis usually involves blood, stool, or urine tests to detect the presence of the bacteria. Early diagnosis is very important because prompt treatment can greatly reduce the severity of the illness. The most effective treatment is antibiotics, which help kill the bacteria causing the infection. However, antibiotic resistance is becoming increasingly common, making treatment more challenging in some regions. Patients are also advised to drink plenty of fluids and rest to support recovery.

Prevention plays a key role in reducing the spread of typhoid fever. Good hygiene practices—such as washing hands before eating and after using the toilet—are essential. Ensuring access to safe drinking water, proper sanitation, and well-cooked food also helps prevent infection. Vaccination is available and recommended for people living in or traveling to high-risk areas.

Although typhoid fever is dangerous, it is largely preventable with the right measures. Improving hygiene, sanitation, and awareness can significantly reduce the number of cases and protect communities from this life-threatening disease.

Dengue Fever

Dengue fever is a viral infection transmitted primarily by the Aedes mosquito, particularly *Aedes aegypti*. It is common in tropical and subtropical regions around the world, especially in areas with warm temperatures, high humidity, and frequent rainfall. Over the past few decades, dengue has become one of the fastest-spreading mosquito-borne diseases, affecting millions of people annually.

3/27
The virus has four distinct serotypes—DENV-1, DENV-2, DENV-3, and DENV-4—which means a person can be infected up to four times. After an infected mosquito bites a person, symptoms typically appear within 4 to 10 days. The most common symptoms include high fever, severe headache, pain behind the eyes, muscle and joint pain, nausea, vomiting, and skin rash. Because of the intense joint and muscle pain, dengue is sometimes referred to as "breakbone fever."

In many cases, dengue fever is mild, and patients recover within a week. However, some individuals may develop severe dengue, also known as dengue hemorrhagic fever or dengue shock syndrome. Severe dengue can cause serious complications such as bleeding, organ damage, and dangerously low blood pressure. Without prompt medical attention, it can be life-threatening.

Diagnosis is made through blood tests that detect the virus or the antibodies produced by the immune system. There is no specific antiviral treatment for dengue, but supportive care is essential. Patients are advised to rest, drink plenty of fluids, and take pain relievers such as paracetamol (acetaminophen). Medications like ibuprofen or aspirin should be avoided because they may increase the risk of bleeding.

Preventing dengue primarily involves controlling mosquito populations and reducing exposure. Removing stagnant water, using mosquito repellents, wearing protective clothing, and installing window screens are effective measures. Community-wide mosquito control efforts also play an important role. A vaccine called Dengvaxia exists, but it is recommended only for individuals who have previously had dengue.

Dengue fever remains a major public-health challenge, but awareness, prevention, and early medical care can significantly reduce its impact.

BCU-14

جراثيم المعدة

Helicobacter pylori (H. Pylori) Infection

يكرهه حموضة الشلح مزمن

Helicobacter pylori, commonly known as H. Pylori, is a spiral-shaped bacterium that infects the stomach lining. It is one of the most widespread chronic infections in the world, especially in developing countries. Many people acquire the infection during childhood, and it can persist for life if left untreated.

بطانة المعدة

The bacteria survive in the stomach by producing enzymes that neutralize stomach acid, allowing them to live in the harsh acidic environment. Over time, H. Pylori can cause chronic inflammation of the stomach lining (gastritis) and may lead to ulcers in the stomach or upper part of the small intestine. In some cases, long-term infection increases the risk of stomach cancer.

Most infected individuals do not show symptoms. However, when symptoms do occur, they may include abdominal pain or burning, bloating, nausea, loss of appetite, frequent burping, and unintentional weight loss. The pain is often worse when the stomach is empty. Because the symptoms are similar to those of other digestive disorders, proper diagnosis is essential.

غثى مبر
انفخاض
انفخاض
تجشؤ متكرر

1-

2-

3-

4-

Several methods are used to diagnose H. Pylori. These include blood tests, stool antigen tests, breath tests that detect bacterial activity, and endoscopy with biopsy. Breath and stool tests are commonly preferred because they are non-invasive and highly accurate.

Treatment for H. Pylori focuses on eradicating the bacteria and healing the stomach lining. The most widely used approach is "triple therapy," which consists of two antibiotics and a proton pump inhibitor (PPI) taken for 10 to 14 days. In regions where antibiotic resistance is high, quadruple therapy may be recommended. Completing the full treatment course is crucial to ensure successful eradication and prevent recurrence.

Prevention strategies include maintaining good hygiene, washing hands properly, drinking clean water, and avoiding contaminated food. Because the exact mode of transmission is not fully understood, improving sanitation and public health measures plays an important role in reducing infection rates.

H. pylori remains a major global health concern due to its complications and high prevalence. With accurate diagnosis, appropriate treatment, and improved hygiene, most infections can be cured, leading to long-term relief and better digestive

Cholera

Cholera is a serious infectious disease caused by the bacterium *Vibrio cholerae*. It mainly affects the small intestine and leads to severe watery diarrhea and dehydration. Cholera is usually transmitted through contaminated water or food, especially in areas with poor sanitation and limited access to clean drinking water. The disease can spread rapidly during natural disasters, wars, or in overcrowded communities.

The main symptom of cholera is sudden onset of profuse watery diarrhea, often described as "rice-water stool" because of its appearance. Other symptoms include vomiting, muscle cramps, rapid heartbeat, low blood pressure, and severe dehydration. If left untreated, cholera can lead to shock and death within a few hours, especially in children and elderly individuals.

Diagnosis of cholera is usually based on clinical symptoms and patient history, particularly in areas where outbreaks are common. Laboratory confirmation can be done by stool culture or rapid diagnostic tests. Early diagnosis is essential to prevent complications and control the spread of the disease.

Treatment of cholera focuses mainly on rapid rehydration. Oral rehydration salts (ORS) are the most important and effective treatment for mild to moderate cases. In severe cases, intravenous fluids are required. Antibiotics may be given to reduce the duration of diarrhea and decrease bacterial shedding, but they are not a substitute for rehydration. Zinc supplements are also recommended for children to reduce the severity and duration of diarrhea.

Prevention of cholera relies on improving water quality, sanitation, and hygiene. Drinking safe water, washing hands with soap, and properly cooking food are key preventive measures. Vaccines against

Preventing malaria involves a combination of strategies. One of the most important methods is reducing mosquito bites. This can be achieved by sleeping under insecticide-treated bed nets, using insect repellents, and wearing long-sleeved clothing, especially during nighttime when *Anopheles* mosquitoes are most active. Environmental measures, such as reducing stagnant water where mosquitoes breed, also play an important role. In addition, some people, such as travelers to malaria-endemic areas, may take preventive medication.

Recent advancements include the development of malaria vaccines such as RTS,S and R21, which provide partial protection, especially for young children. While these vaccines are an important step forward, they are used in addition to—not instead of—other prevention measures.

Overall, malaria remains a major but preventable disease. Through improved diagnosis, effective treatment, and community-wide prevention efforts, the global burden of malaria can be significantly reduced.

Diagnosis of diabetes is made through blood tests such as fasting blood glucose, random blood glucose, HbA1c test, and oral glucose tolerance test. Early diagnosis is important to prevent long-term complications.

Management of diabetes involves lifestyle modifications and medications. A healthy diet, regular physical activity, weight control, and blood glucose monitoring are essential. Medications include oral antidiabetic drugs and insulin injections, depending on the type and severity of the disease. Patient education plays a vital role in successful diabetes management and improving quality of life.

Hypertension (High Blood Pressure)

Hypertension, commonly known as high blood pressure, is a chronic condition in which the force of blood against the walls of the arteries is consistently too high. It is one of the most common cardiovascular diseases worldwide and is often called the "silent killer" because it usually has no noticeable symptoms.

Blood pressure is measured in millimeters of mercury (mmHg) and is recorded as systolic pressure over diastolic pressure. Normal blood pressure is around 120/80 mmHg. Hypertension is generally diagnosed when blood pressure readings are consistently above 140/90 mmHg.

Risk factors for hypertension include aging, family history, obesity, excessive salt intake, lack of physical activity, smoking, stress, and excessive alcohol consumption. Although many people with hypertension do not experience symptoms, some may suffer from headaches, dizziness, blurred vision, or nosebleeds in severe cases.

Diagnosis is made through repeated blood pressure measurements using a sphygmomanometer. Additional tests may be performed to assess organ damage, such as blood tests, urine tests, ECG, and kidney function tests.

Treatment of hypertension aims to reduce blood pressure and prevent complications such as heart attack, stroke, kidney disease, and heart failure. Lifestyle changes are the first line of management and include a balanced diet, reduced salt intake, regular exercise, weight loss, and stress management. When lifestyle modifications are not sufficient, antihypertensive medications are prescribed. Regular monitoring and adherence to treatment are essential for effective blood pressure control.

Grammar

Subject and Predicate

What Is a Sentence?

A sentence is a group of words that expresses a complete idea.

Every complete sentence has two main parts:

Subject

Predicate

The Subject

The subject tells who or what the sentence is about.

Examples:

The student reads a book.

My brother is a doctor.

The red car stopped suddenly.

Types of Subjects :-

1. Simple Subject

The main noun or pronoun in the subject sentence.

The teacher explained the lesson.

The dog barked loudly.

2. Complete Subject

The simple subject plus all words that describe it.

The new teacher explained the lesson.

The small brown dog barked loudly.

5- The teacher explained the rule.

Exercise 2:

Identify the Predicate

Underline the predicate in each sentence.

1- The baby is sleeping peacefully.

2- The bus arrived late.

3- My sister is studying medicine.

4- The birds flew away.

5- The students completed the test.

Exercise 3:

Match each subject with the correct predicate.

Subjects

- The doctor
- The cat
- My parents
- The rain

Predicates

- Are traveling tomorrow
- is sleeping on the sofa
- treated the patient
- stopped suddenly

Exercise 4:

Complete the Sentence

Add a suitable subject or predicate.

- a) _____ is reading a novel.
- b) The little girl _____.
- c) _____ are playing football.
- d) The movie _____.
- e) _____ won the competition.

Exercise 5:

Write Your Own Sentences

Write three sentences.

Underline the subject once and the predicate twice.

Types of Sentences

Exercise 1:

Identify the Type

Write Declarative (D), Interrogative (I), Imperative (Imp), or Exclamatory (E).

- a) Where are my keys?
- b) Please open the door.
- c) The sky is very clear today.
- d) How exciting this trip is!
- e) Did you finish your assignment?
- f) Wash your hands before eating.
- g) I love reading books.
- h) What a wonderful performance!

Exercise 2:

Rewrite Sentences

- a) I am tired. → Make it interrogative.
- b) Close the door. → Make it exclamatory.
- c) She is a good singer. → Make it imperative.
- d) What a surprise! → Make it declarative.

Exercise 3:

Complete the sentences according to the type in brackets.

- a) _____ ! I can't believe my eyes. (Exclamatory)
- b) _____ the lights, please. (Imperative)
- c) _____ going to the party tonight? (Interrogative)
- d) _____ is my favorite subject. (Declarative)

Sentence Pattern

What Is a Sentence Pattern?

A sentence pattern is the structure of a sentence, showing how the subject, verb, and other elements are arranged.

Understanding sentence patterns helps you write and speak correctly in English.

Basic Sentence Patterns

Pattern 1:

Subject + Verb (S + V)

The sentence has a subject and an action (verb).

Examples:

- Birds fly.
- He sleeps.

Pattern 2:

The Predicate

The predicate tells what the subject does or what happens to the subject.

Examples:

The student reads a book.

My brother is a doctor.

The red car stopped suddenly.

Types of Predicates —

1. Simple Predicate

The main verb in the sentence.

She runs fast.

They finished the work.

2. Complete Predicate

The verb plus all words that come after it.

She runs very fast every morning.

They finished the work on time.

Identifying Subject and Predicate

Example Sentence:

The young boy is playing in the park.

Subject: The young boy

Predicate: is playing in the park

Important Notes

A sentence cannot exist without both a subject and a predicate.

The subject usually comes before the predicate.

Commands often have an implied subject.

Example:

Close the door.

Subject: You (implied)

Predicate: close the door

Exercises

Exercise 1:

Identify the Subject

Underline the subject in each sentence.

1- The tall man opened the door.

2- My best friend lives nearby.

3- The children are playing outside.

4- A beautiful painting hangs on the wall.

Subject + Verb + Object (S + V + O)

The sentence has a subject, a verb, and an object receiving the action.

Examples:

- She reads a book.
- I like chocolate.

Pattern 3:

Subject + Verb + Complement (S + V + C)

The sentence has a subject, a verb, and a complement that describes or renames the subject.

Examples:

- He is a teacher.
- The weather feels cold.

Types of Complements:

1. **Subject complement** (after linking verbs like be, seem, become)
2. **Object complement** (after some verbs: consider, make, call)

Pattern 4:

Subject + Verb + Indirect Object + Direct Object (S + V + IO + DO)

The sentence has a subject, verb, and two objects:

Indirect Object (IO): receives the direct object

Direct Object (DO): the thing being given or acted on

Examples:

- She gave me a gift.
- The teacher taught the students a lesson.

Pattern 5:

Subject + Verb + Object + Complement (S + V + O + C)

The sentence has a subject, a verb, an object, and a complement that describes the object.

Examples:

- They elected him president.
- We found the movie interesting.

Pattern 6:

Subject + Verb + Adverb (S + V + Adv)

The sentence has a subject, a verb, and an adverb or adverbial phrase.

Examples:

- He runs quickly.
- They arrived yesterday.

• Notes on Sentence Patterns

- Some verbs are linking verbs (be, seem, become) and connect the subject with a complement.
- Action verbs usually have an object or adverb.
- Sentence patterns help identify errors in writing or speaking.

Exercises

Exercise 1:

Write the pattern number for each sentence.

- a. The cat sleeps.
- b. She plays the piano.
- c. He is my brother.
- d. The teacher gave the students homework.
- e. They found the test easy.
- f. Birds sing beautifully.

Exercise 2:

Fill in the blanks with suitable words.

- a. She _____ a letter to her friend. (S + V + IO + DO)
- b. The flowers _____ very colorful. (S + V + C)
- c. He _____ quickly to the station. (S + V + Adv)
- d. We _____ the room clean. (S + V + O + C)
- e. Dogs _____ loudly at night. (S + V)

Exercise 3:

Write one sentence for each pattern.

Use underline for subject once and predicate twice.

- a. S + V
- b. S + V + O
- c. S + V + C
- d. S + V + IO + DO
- e. S + V + O + C
- f. S + V + Adv

Exercise 4:

Underline the subject once and predicate twice in each sentence.

- a. The children played football in the park.
- b. She seems tired after work.
- c. I sent my friend a postcard.
- d. We consider him a genius.
- e. He walks slowly to school.

Types of Sentences

Types of Sentences by Structure

- Sentences can be simple, compound, complex, or compound-complex.

1- Simple Sentence

Contains one independent clause (a subject and a predicate).

Example:

- I like apples.
- She reads every day.

2- Compound Sentence

Contains two or more independent clauses joined by a coordinating conjunction (for, and, nor, but, or, yet, so — FANBOYS).

Example:

- I like apples, and she likes oranges.
- He was tired, but he continued working.

3- Complex Sentence

Contains one independent clause and at least one dependent (subordinate) clause.

Dependent clauses often start with because, although, if, when, since, while.

Example:

- I stayed home because it was raining.
- She smiled when she saw the gift.

4- Compound-Complex Sentence

Contains at least two independent clauses and at least one dependent clause.

Example:

- I stayed home because it was raining, and my brother watched a movie.
- She likes coffee, but she avoids it when she is stressed.

Types of Sentences by Function

Sentences can also be categorized by their purpose:

1- Declarative Sentence

Makes a statement.

Ends with a period (.)

Example:

- I am learning English.
- The sky is blue.

2- Interrogative Sentence

Asks a question.

Ends with a question mark (?).

Example:

- Are you coming to the party?
- What time is it?

3- Imperative Sentence

Gives a command or request.

Ends with a period (.) or exclamation mark (!)

Example:

- Close the door.
- Please help me with this task.

4- Exclamatory Sentence

Shows strong emotion or excitement.

Ends with an exclamation mark (!)

Example:

- What a beautiful day!
- I can't believe we won

Phrasal Verbs

What are Phrasal Verbs?

A phrasal verb is a combination of:

A verb + a preposition or adverb

Together, they create a new meaning different from the original verb

Examples:

- Look after – to take care of
She looks after her little brother.
- Turn on – to switch on
Please turn on the lights.
- Give up – to stop trying
He gave up smoking last year.

Types of Phrasal Verbs

1. Transitive Phrasal Verbs (needs an object)

- Bring up – to mention
She brought up an important topic.
- Call off – to cancel

What Is a Sentence?

A sentence is a group of words that expresses a complete thought.

Sentences are classified into four main types based on their purpose or meaning.

The Four Types of Sentences

6

1. Declarative Sentences (Statements)

A declarative sentence gives information or makes a statement.

It ends with a period (.).

Examples:

I like chocolate.

The sun rises in the east.

She is studying for her exam.

2. Interrogative Sentences (Questions)

An interrogative sentence asks a question.

It ends with a question mark (?).

Examples:

Are you ready?

What is your name?

Did they finish the homework?

3. Imperative Sentences (Commands or Requests)

An imperative sentence gives a command, instruction, or request.

It usually ends with a period (.) but can also use an exclamation mark (!).

The subject is often implied as "you".

Examples:

Close the window.

Please sit down.

Listen carefully!

4. Exclamatory Sentences (Strong Emotions)

An exclamatory sentence expresses strong feelings like surprise, anger, joy, or excitement.

It ends with an exclamation mark (!).

Examples:

What a beautiful day!

I can't believe it!

How amazing this movie is!

How to Identify Sentence Types

- Type
- Question to Ask
- Ending Punctuation
- Declarative (Does it give information)
- Interrogative (Does it ask a question)
- Imperative (Does it give a command or request)
- Exclamatory (Does it show strong emotion)

Exercises

Look after	c) Collect
Give up	d) Quit

Exercise 3:

Write a sentence for each phrasal verb:

1. Turn off
2. Wake up
3. Call off

Capitalization and Punctuation

Capitalization (Using Capital Letters)

Capital letters are used in specific places:

1- The First Word of a Sentence

Every sentence begins with a capital letter.

Example:

- She is reading a book.
- Today is Monday.

2- Names of People

Capitalize first and last names.

Example:

- John Smith
- Mary Johnson

3- Names of Places

Countries, cities, streets, landmarks.

Example:

London, Egypt, Nile River

4- Days, Months, and Holidays

Example:

Monday, January, Christmas

5- Titles

Titles of books, movies, songs, articles.

Example:

- Harry Potter and the Sorcerer's Stone
- The Lion King

6- Pronoun "I"

They called off the meeting.

2. Intransitive Phrasal Verbs (no object needed)

- Break down – stop working (machine)
My car broke down yesterday.
- Wake up – stop sleeping
I usually wake up at 7 a.m.

3. Separable vs. Inseparable Phrasal Verbs

Separable: object can go between verb and particle

- Turn off the TV → Turn the TV off ✓

Inseparable: object always after verb + particle

- Look after children (not: Look children after ✗)

• Common Phrasal Verbs

Phrasal Verb	Meaning	Example
Look after	Take care of	I look after my dog.
Look for	Search	She is looking for her keys.
Give up	Quit	He gave up smoking.
Turn on	Switch on	Turn on the computer.
Turn off	Switch off	Please turn off the lights.
Call off	Cancel	They called off the event.
Get up	Rise from bed	I get up at 6 a.m.
Break down	Stop working	The car broke down yesterday.
Run out of	Finish supply	We ran out of milk.
Pick up	Collect	Can you pick up the package?

Exercises

Exercise 1:

Fill in the blanks

Please _____ the lights, it's too dark. (turn on / turn off)
I can't _____ my old toys. (give up / look after)
We had to _____ the meeting because of the storm. (call off / pick up)
She _____ her little brother while her parents were away. (look after / run out of)
My phone _____ yesterday and I couldn't call anyone. (broke down / get up)

Exercise 2:

Match the phrasal verbs with the meaning

Phrasal Verb	Meaning
Run out of	a) Take care of
Pick up	b) Finish supply

Colon (:) introduces a list or explanation:

Example:

- I need three things: a pen, a notebook, and an eraser.

8- Semicolon (;) connects related sentences:

Example:

- I went to the park; it was very crowded.

Exercises

Exercise 1:

Rewrite the sentences with correct capitalization:

- My friend john is from paris.
- Christmas is celebrated in december.
- I like the movie the lion king.
- Microsoft is a big company.
- Yesterday was Monday.

Exercise 2:

Add the correct punctuation to these sentences:

- What time is it
- I bought apples oranges and bananas
- Watch out
- She said I will be late
- I went to the park it was raining

Exercise 3:

Correct capitalization and punctuation in this paragraph:

Yesterday my brother and i went to London we visited the British museum it was amazing we took lots of pictures after that we went to a restaurant called the red lion

English Tenses

What are Tenses?

Tenses tell us when an action happens. There are three main time frames:

- Present – action happening now or repeatedly
- Past – action that happened before now
- Future – action that will happen later

Each time frame has four forms: Simple, Continuous, Perfect, and Perfect Continuous.

1- Present Tenses

Present Simple

Always capitalized.

Example:

- I am learning English.

7- Organizations and Brand Names

Example:

United Nations, Microsoft

Punctuation (Using Marks Correctly)

1- Period (.)

Used at the end of a declarative sentence.

Example:

- She likes chocolate.

2- Question Mark (?)

Used at the end of a question.

Example:

- Where are you going?

3- Exclamation Mark (!)

Shows strong feeling or surprise.

Example:

- Watch out!
- I am so happy!

4- Comma (,)

Separate items in a list:

Example:

- I bought apples, bananas, and oranges.

After introductory words or phrases:

Example:

- After lunch, we went for a walk.

5- Apostrophe (')

Show possession:

Example:

- Mary's book

Contractions:

Example:

- don't = do not, I'm = I am

6- Quotation Marks (" ")

To show someone's words:

Example:

- She said, "I love reading."

7- Colon (:) and Semicolon (;)

used for habits, general truths, and repeated actions.
Example:

- She reads every day.

Present Continuous

used for actions happening now or around the present time.
Example:

- She is reading a book right now.

Present Perfect

used for actions completed at an unspecified time, or with results relevant now.
Example:

- She has read three books this week.

Present Perfect Continuous

used for actions that started in the past and are still continuing.
Example:

- She has been reading for two hours.

2- Past Tenses

Past Simple

used for actions completed in the past at a specific time.
Example:

- He played football yesterday.

Past Continuous

used for actions in progress at a specific moment in the past.
Example:

- He was playing football at 5 p.m.

Past Perfect

used for actions completed before another past action.
Example: He had played football before it rained.

Past Perfect Continuous

used for actions that were in progress before another past action.
Example:

- He had been playing for two hours before it rained.

3- Future Tenses

Future Simple

used for decisions, predictions, or promises.
Example:

- She will go to the park tomorrow.

Going to Future

used for planned actions or intentions.
Example:

- She is going to visit her friend.

Future Continuous

used for actions that will be in progress at a specific future time.
Example:

- She will be studying at 6 p.m.

Future Perfect

used for actions that will be completed before a specific future time.

Example:

- She will have finished her homework by 8 p.m.

4- Positive, Negative, and Question Forms

Example	Positive	Negative	Question
Present Simple	She reads books.	She does not (doesn't) read books.	Does she read books?
Present Continuous	She is reading a book.	She is not (isn't) reading a book.	Is she reading a book?

Tip:

Practice forming positive, negative, and question forms for all tenses.

Exercises

Exercise 1:

Choose the correct tense

- She (reads / is reading) every morning.
- They (played / had played) football before it rained.
- I (will go / am going to go) to the market tomorrow.
- He (has been working / worked) for three hours.
- We (were watching / watched) TV at 8 p.m. yesterday.

Exercise 2:

Complete the sentences

- She _____ (study) English for two years. (Present Perfect Continuous)
- I _____ (finish) my homework before dinner. (Past Perfect)
- They _____ (visit) Paris next summer. (Future Simple)
- He _____ (play) football when I called him. (Past Continuous)
- We _____ (meet) our friends yesterday. (Past Simple)

Exercise 3:

Make questions

- She is reading a book. → _____?
- They have finished their work. → _____?
- He will go to the park. → _____?
- You were watching TV. → _____?
- I am going to visit my friend. → _____?

Exercise 4:

Correct the sentences

- He go to school every day.
- I am study English now.
- They has finished their homework.
- She will going to the party tomorrow.
- We was playing football yesterday.

=====

Name:

Facility:

=====

Question one:

Give the Arabic name for the following diseases.

1. Multiple sclerosis
2. Epilepsy
3. Parkinson's disease
4. Alzheimer's disease
5. Meningitis
6. Anemia
7. Leukemia
8. Hemophilia
9. Thalassemia
10. Sickle cell disease
11. Acute kidney injury
12. Chronic kidney disease
13. Kidney stones
14. Nephrotic syndrome
15. Polycystic kidney disease
16. Depression
17. Anxiety disorder
18. Bipolar disorder
19. Schizophrenia
20. Obsessive-compulsive disorder
21. Hepatitis A
22. Hepatitis B
23. Hepatitis C



24. Liver cirrhosis
25. Fatty liver disease
26. Liver cancer
27. Neuropathy
28. Autoimmune encephalitis
29. Hemolytic anemia
30. Glomerulonephritis

Question Two:

In not less than 150 words write about the pharmaceutical industry in Sudan.

Question Three:

In not less than 150 words write about the misuses of medication and its consequences.

Question Four:

In not less than 100 words express yourself observing the grammatical rules and punctuation marks

=====Good luck=====